VxLEARN Networks

Networking & Cybersecurity Track
Simulated Employment Program

Lab Report:
Wireless Router Hardening

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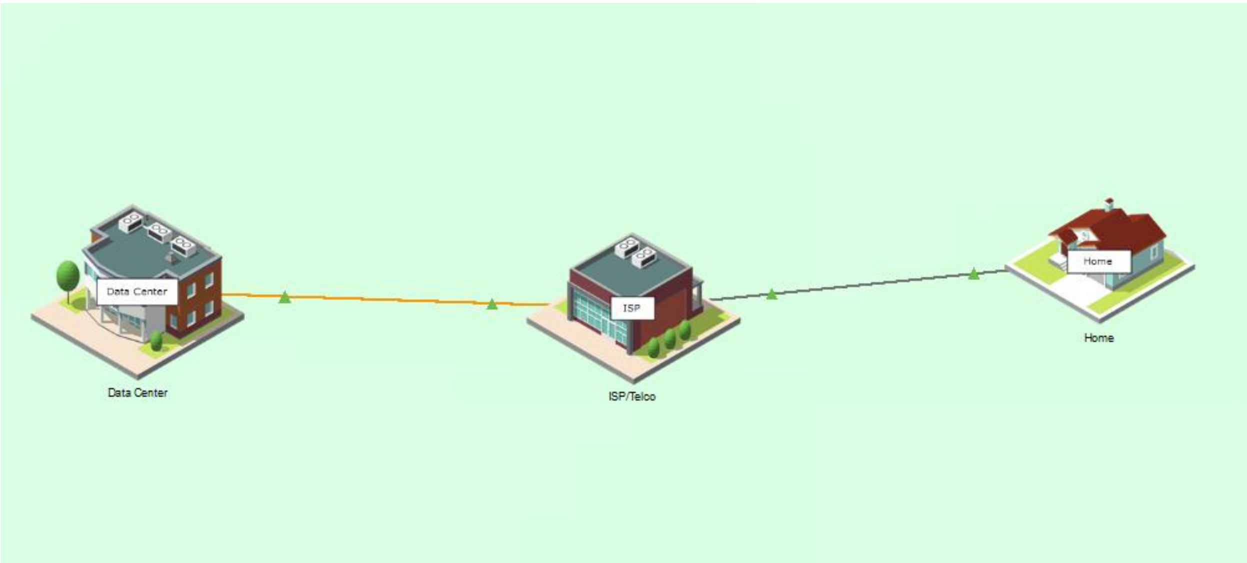
Objectives

- Part 1: Configure Basic Security Settings for a Wireless Router
- Part 2: Configure Wireless Router Network Security
- Part 3: Configure Wireless Clients Network Security
- Part 4: Verify Connectivity and Security Settings

Background / Scenario

In this Packet Tracer activity, a wireless router was partially configured but has security issues that need to be addressed. The router will be hardened to mitigate potential attacks, wireless networks will be secured, wireless clients will be connected, and connectivity and security settings will be verified.



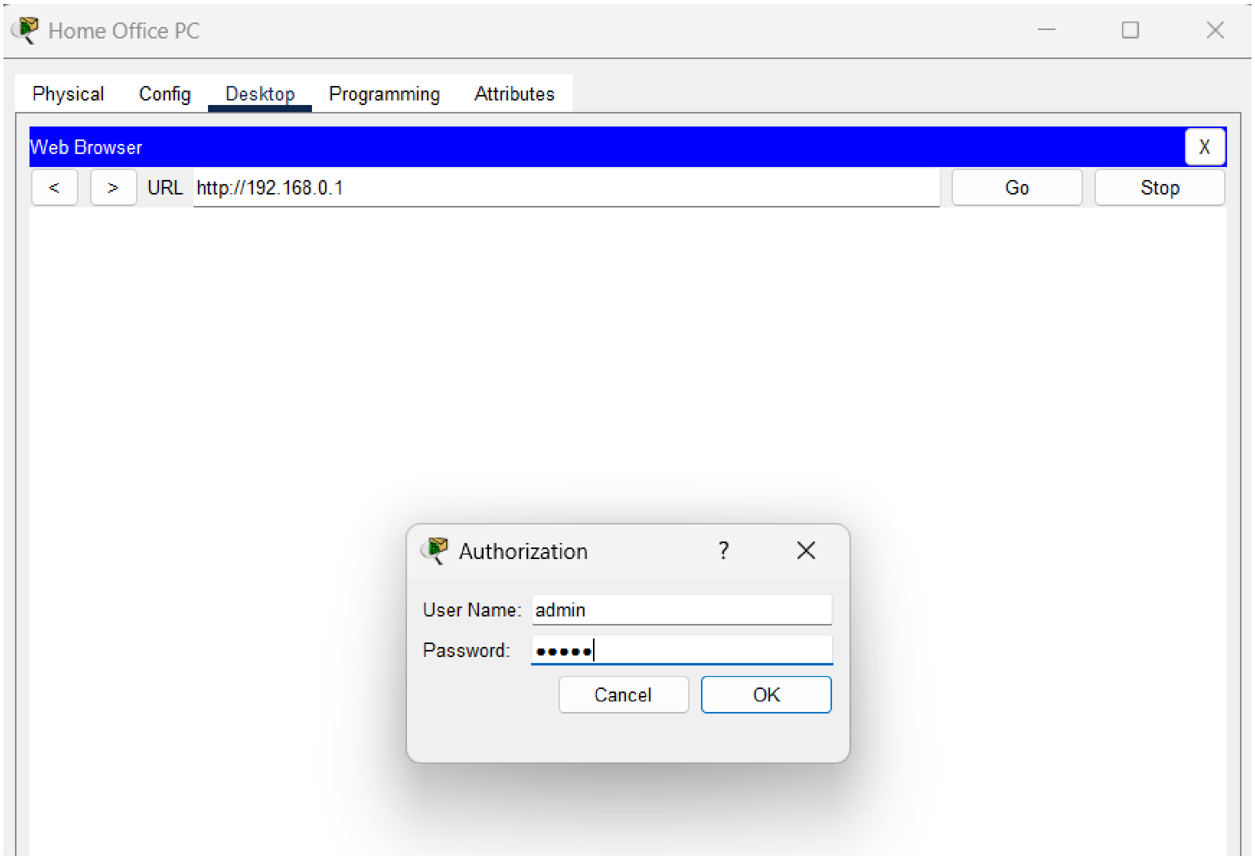
Part 1: Configure Basic Security Settings for a Wireless Router

The initial settings on a home wireless router can pose a security risk. In this part, the router password is changed and remote login is disabled.

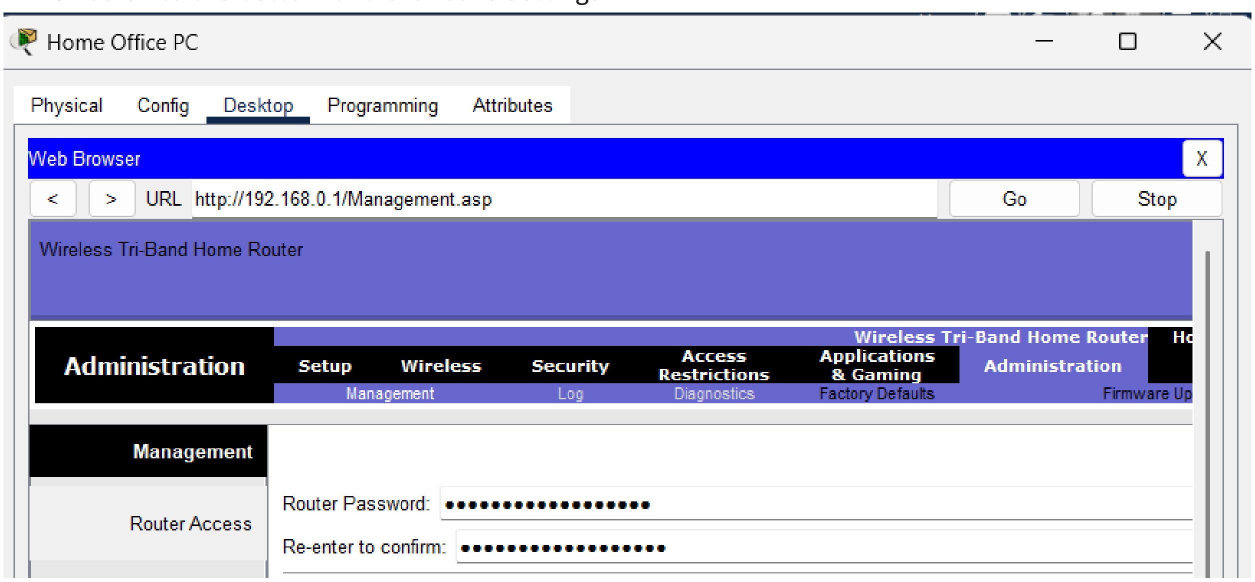
Step 1: Change the Default Router Password

A strong password or passphrase prevents unauthorised users from changing router configurations.

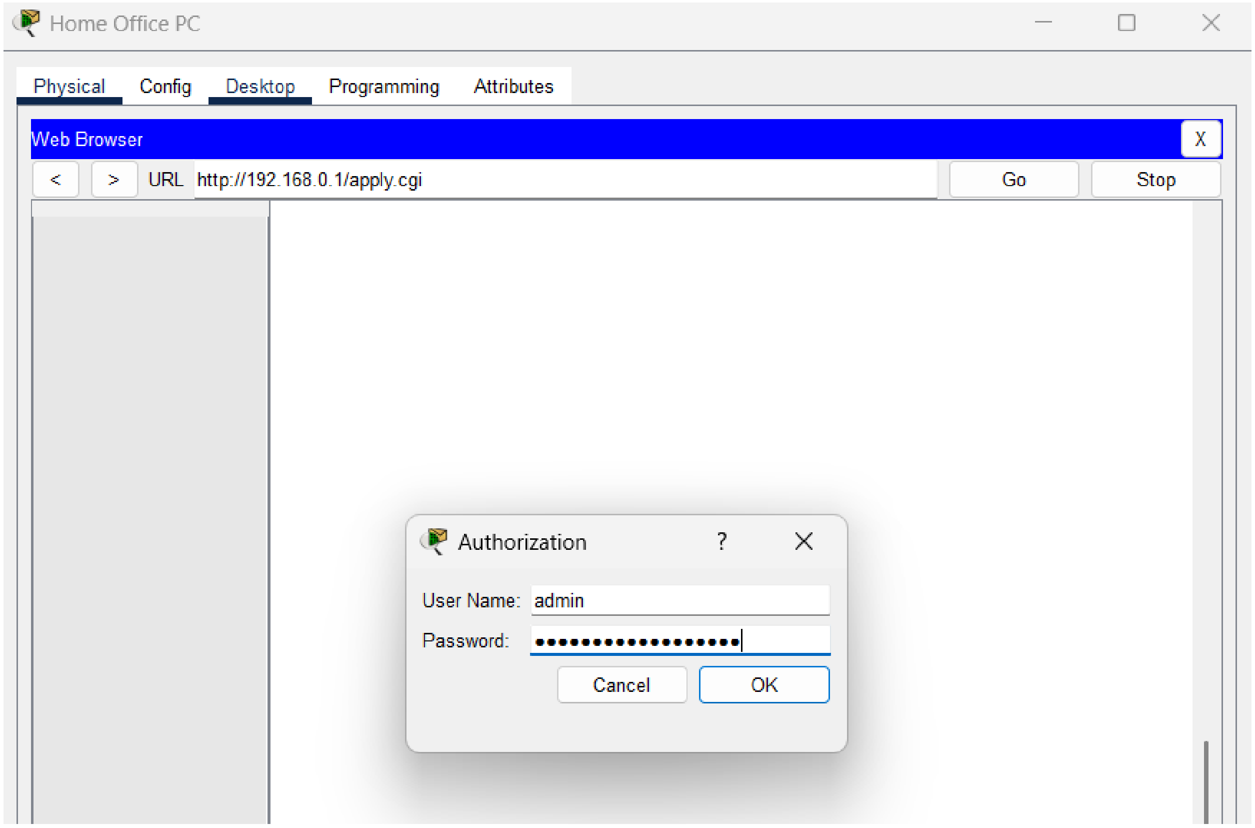
- **a.** Click Home, then Home Office PC > Desktop tab > Web Browser.
- **b.** Connect to the Home Wireless Router at 192.168.0.1.
- **c.** Enter admin for both the username and password.

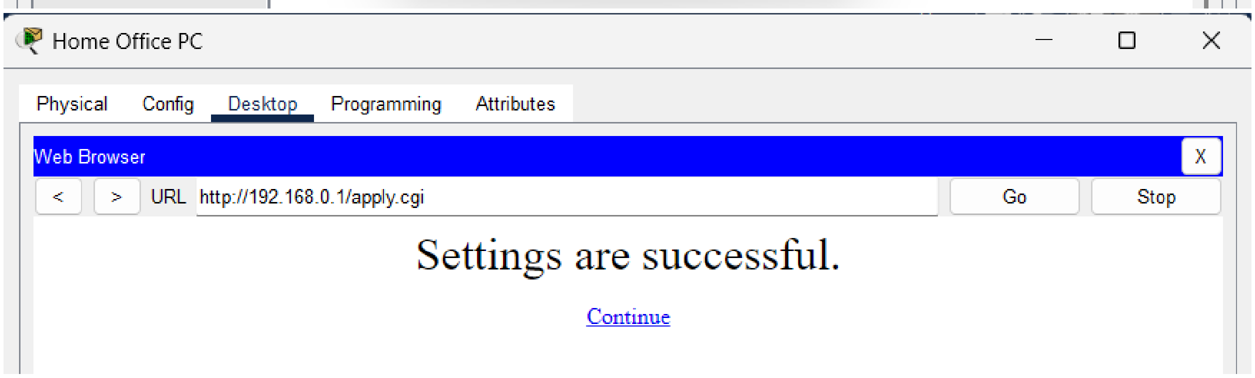


- **d.** Click the Administration tab, then enter cisconetacadrocks! for both password fields.
- **e.** Scroll to the bottom and click Save Settings.



- **f.** The router requests re-authentication. Log back in with username admin and the new password, then click Continue.

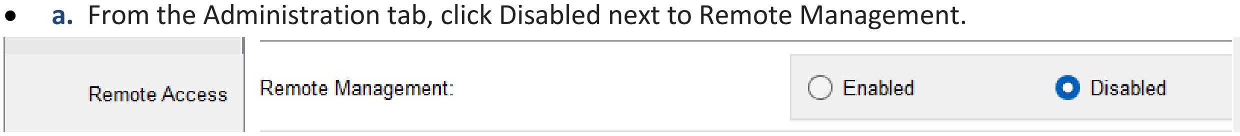




Step 2: Disable Remote Management

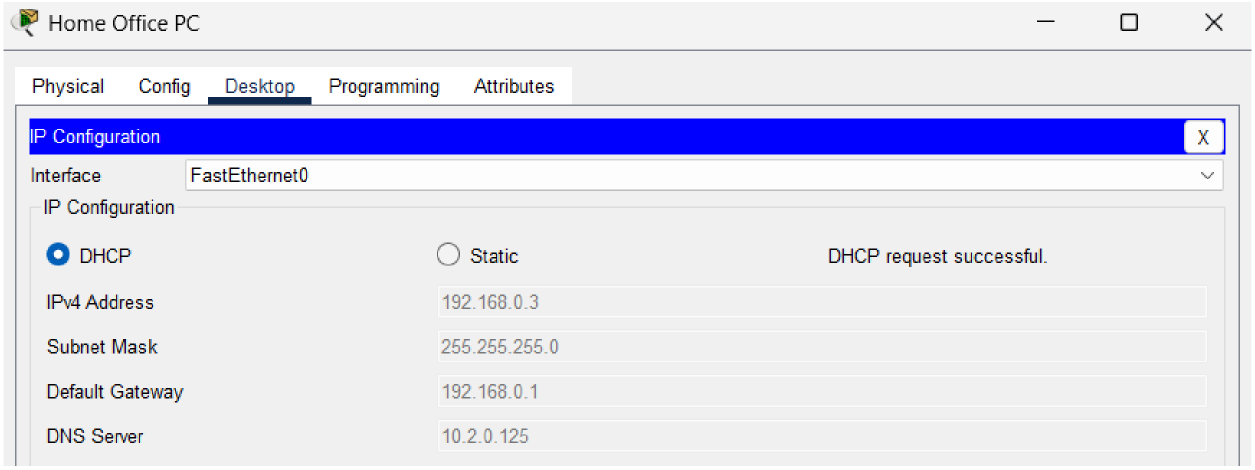
Remote management requires an open port, which can be exploited by attackers. Disabling it removes this attack vector.

- **a.** From the Administration tab, click Disabled next to Remote Management.



- **b.** Click Save Settings.

> **Note:** *This change will cause the router to reset. Close the Web Browser and click Fast Forward Time (Alt+D). Return to Home Office PC and check IP Configuration. Toggle between DHCP and Static if necessary until Home Office PC receives an address from the 192.168.0.1/24 network. Then re-open the Web Browser and navigate back to 192.168.0.1 to re-authenticate.*



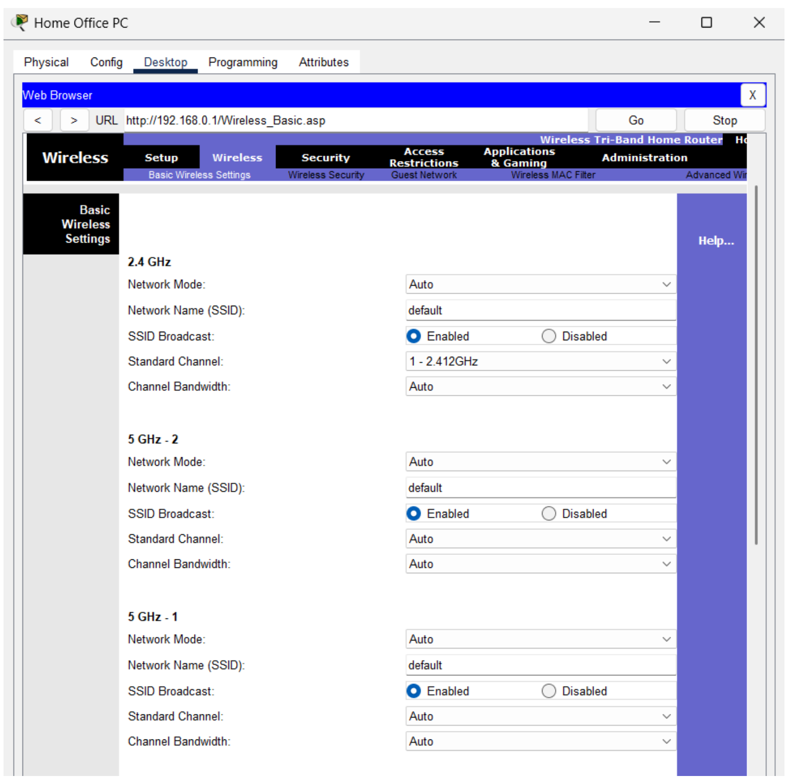
Part 2: Configure Wireless Router Network Security

In this part, the wireless networks are secured so that only devices with the correct configuration can connect.

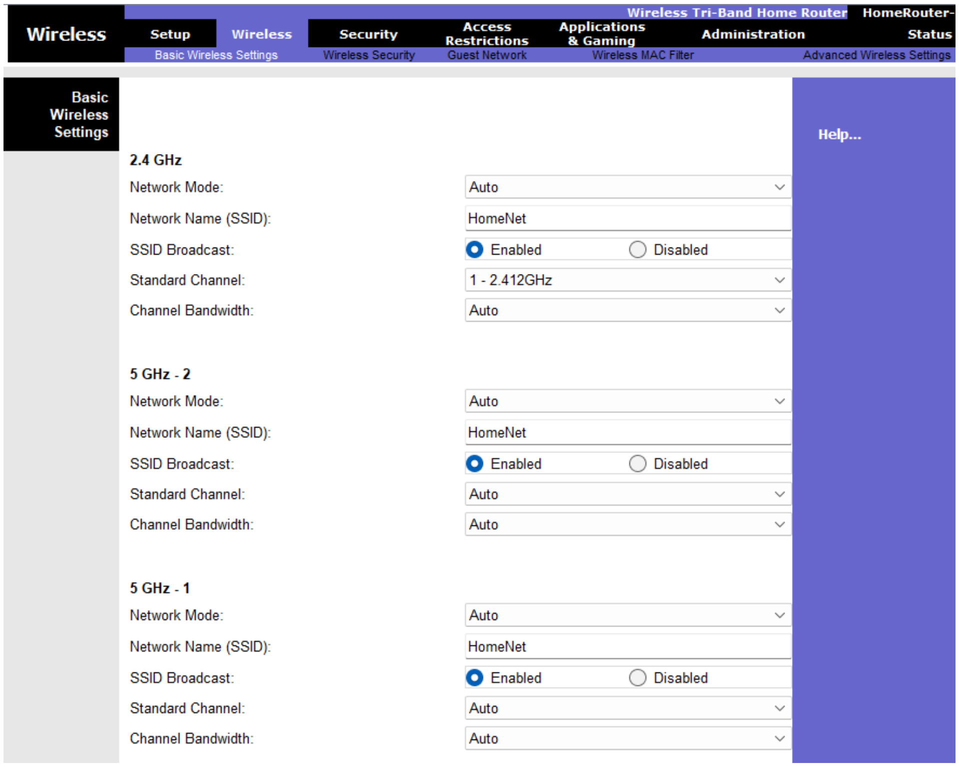
Step 1: Configure and Broadcast the HomeNet SSID

Broadcasting a clear SSID name allows legitimate users to identify and connect to the correct network.

- **a.** Click the Wireless tab. For each of the three networks, click Enabled for SSID Broadcast.



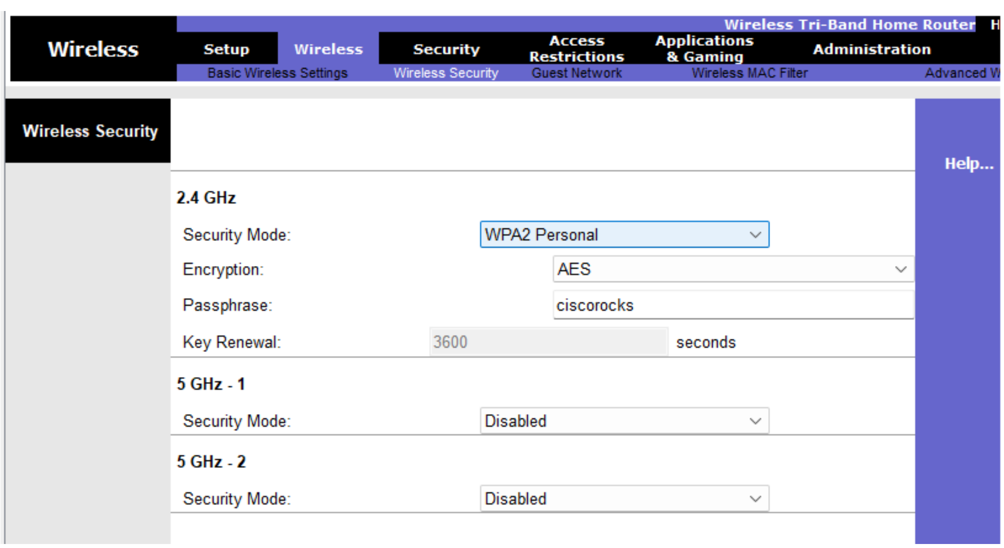
- **b.** For each of the three networks, change the SSID from default to HomeNet.
- **c.** Click Save Settings.



Step 2: Configure Security for the HomeNet Wireless Networks

Encrypting wireless traffic prevents eavesdropping and traffic interception by attackers.

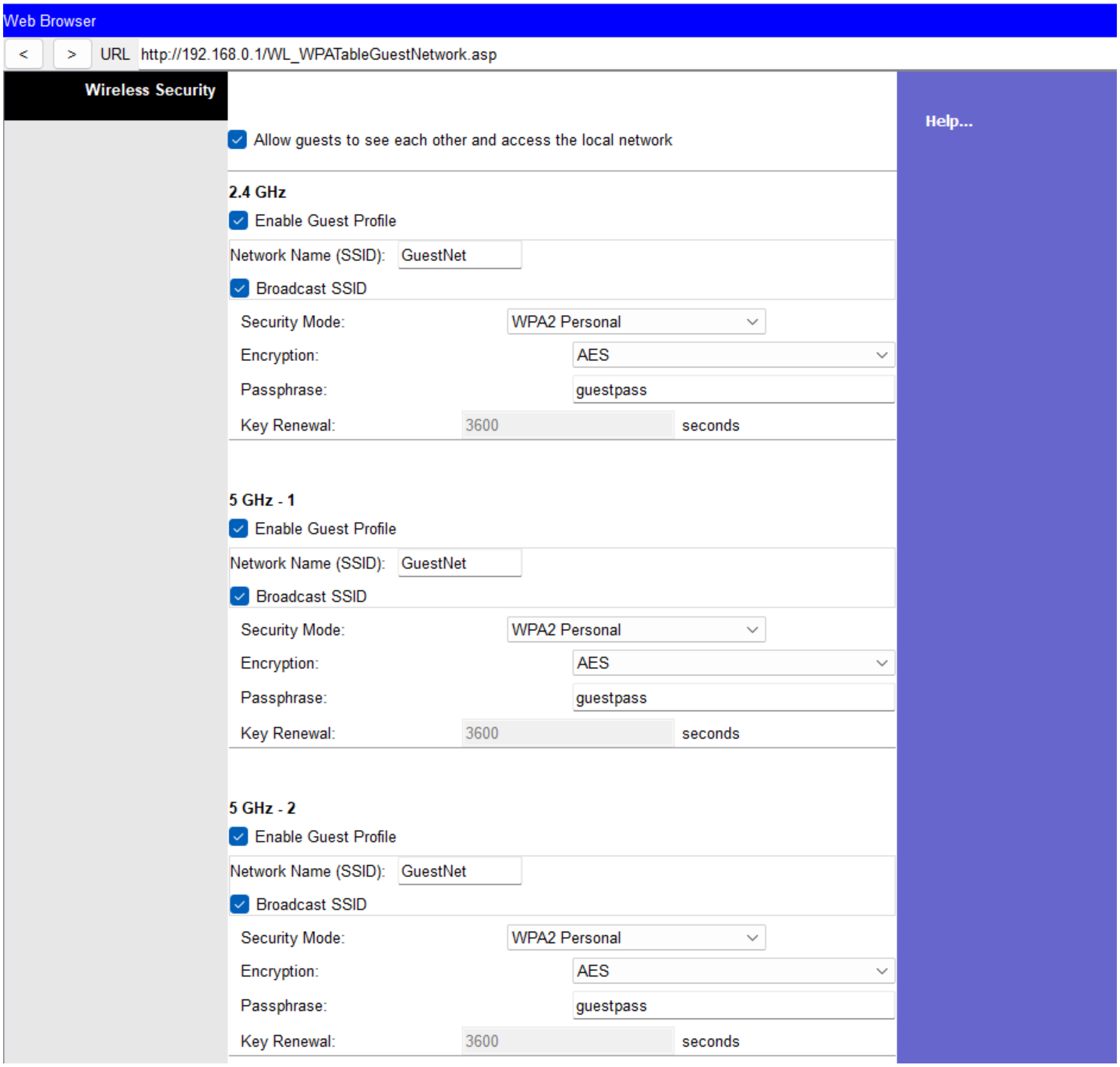
- **a.** From the Wireless tab, click Wireless Security.
- **b.** For each of the three networks, set: Security Mode: WPA2 Personal | Encryption: AES | Passphrase: ciscorocks
- **c.** Click Save Settings.



Step 3: Configure Security for the GuestNet Wireless Network

A separate GuestNet keeps visitor traffic isolated from the main home network.

- **a.** From the Wireless tab, click Guest Network.
- **b.** For each of the three networks, click Enable Guest Profile and configure: SSID: GuestNet | Broadcast SSID: Enabled | Security Mode: WPA2 Personal | Encryption: AES | Passphrase: guestpass
- **c.** Click Save Settings.

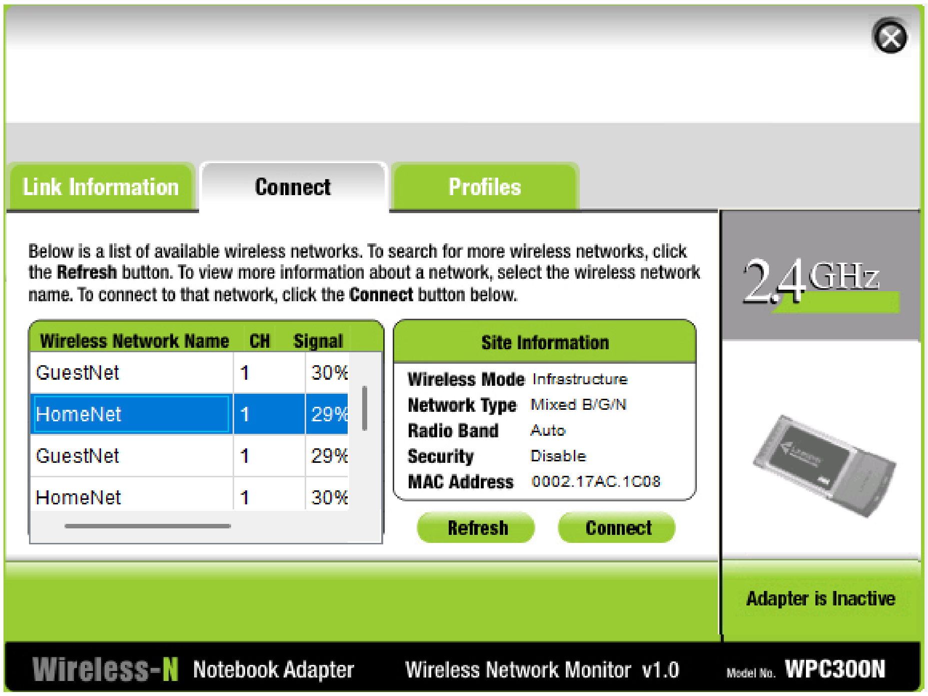


Part 3: Configure Wireless Clients Network Security

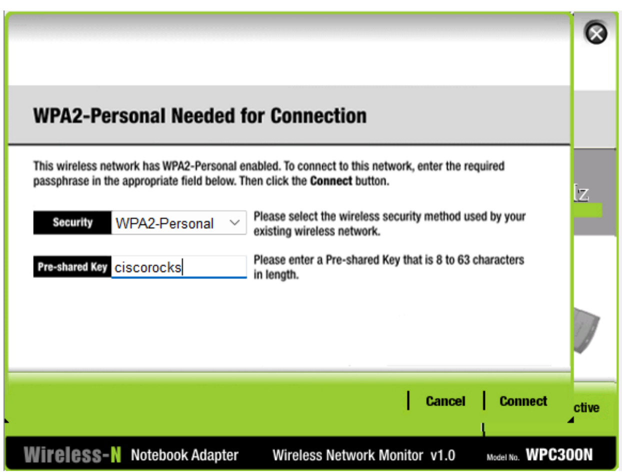
In this part, wireless clients are connected to the secured networks using the correct credentials.

Step 1: Configure Wireless Connectivity for the Laptops

- **a.** Click Home Laptop 1 (living room) > Desktop tab > PC Wireless.
- **b.** Click the Connect tab.
- **c.** Click the first entry for HomeNet, then click Connect.

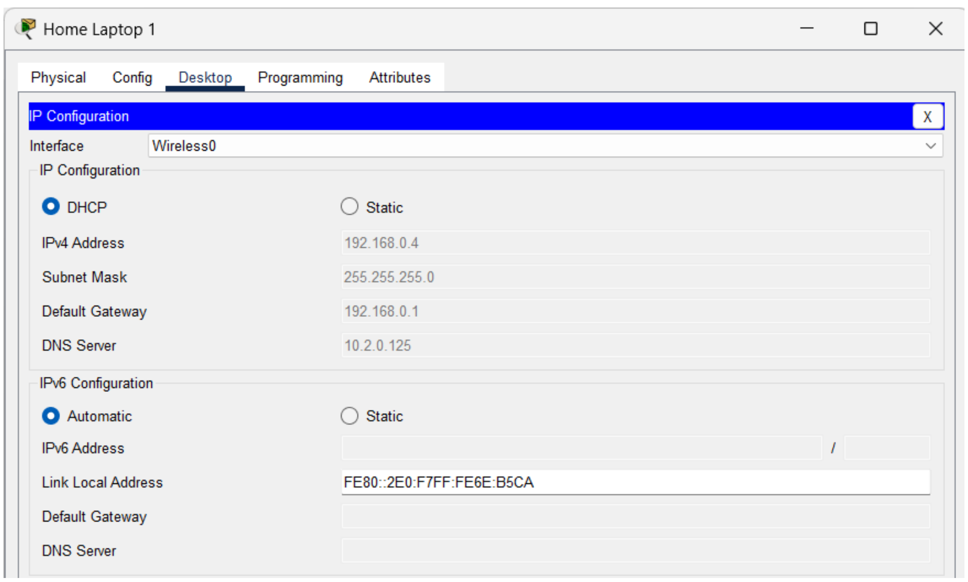


- **d.** Security is already set to WPA2-Personal. Enter ciscorocks for the Pre-shared Key, then click Connect.

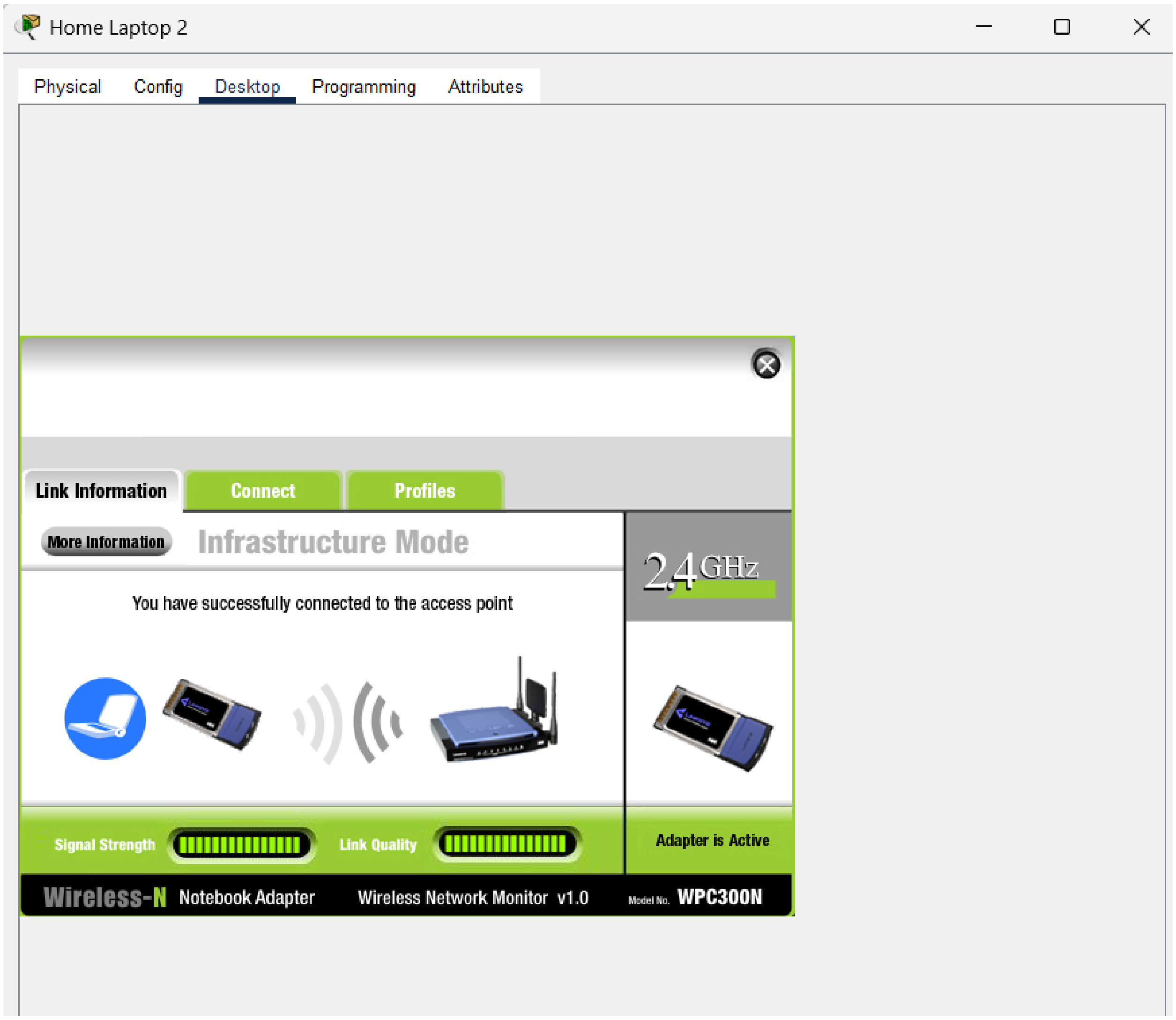


- **e.** Click the Link Information tab. Confirm the message: "You have successfully connected to the access point".

- **f.** Close PC Wireless and click IP Configuration. Toggle between DHCP and Static if needed until the laptop receives an address from the 192.168.0.0/24 network.

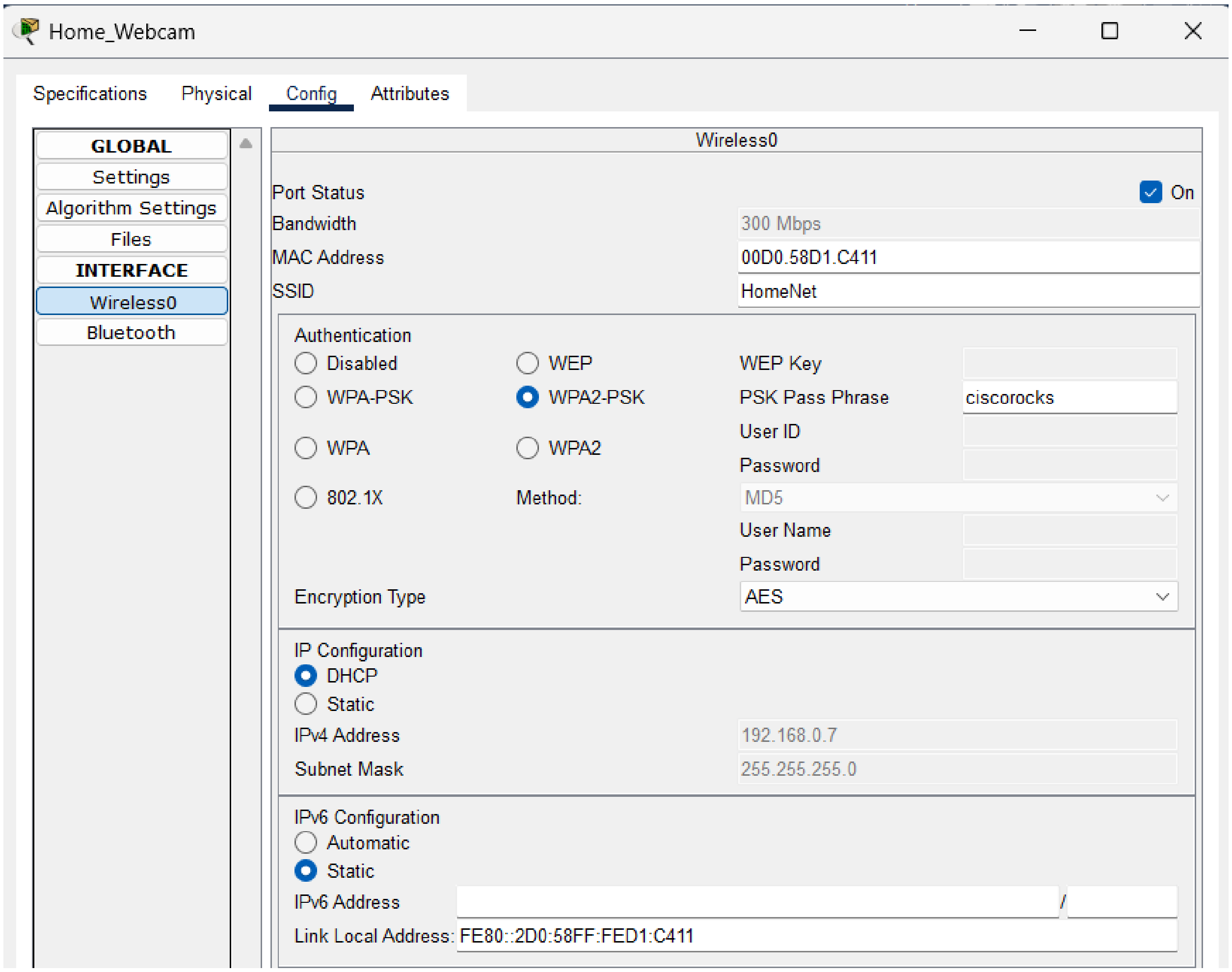


- **g.** Repeat this step for Home Laptop 2 (home office) but connect to GuestNet using passphrase guestpass.


Home Laptop 2
Physical
Config
Desktop
Programming
Attributes
Link Information
Connect
Profiles
More Information
Infrastructure Mode
2.4 GHz
You have successfully connected to the access point
Signal Strength
Link Quality
Adapter is Active
Wireless-N Notebook Adapter
Wireless Network Monitor v1.0
Model No. WPC300N

Step 2: Configure Wireless Connectivity for the IoT Devices

- **a.** Click Home_Webcam (top shelf of bookcase in home office) > Config > Wireless0.
- **b.** Enter HomeNet for the SSID.
- **c.** Choose WPA2-PSK for Authentication, then enter ciscorocks for the PSK Pass Phrase.
- **d.** Under IP Configuration, click DHCP. Verify the device received an address from the 192.168.0.0/24 network. Toggle between DHCP and Static if necessary.
- **e.** Repeat this step for Home_Siren (living room, above bookcase) and Home Doors.

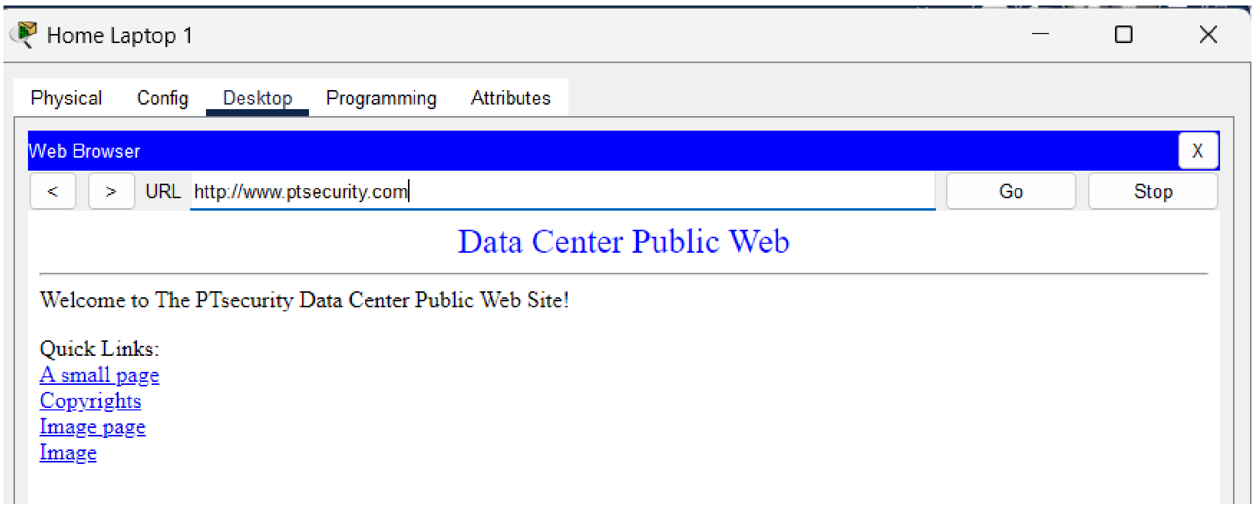


Part 4: Verify Connectivity and Security Settings

In this part, configurations and security settings are tested to confirm that devices are communicating correctly and that network isolation is working.

Step 1: Test Internet Connectivity for Wireless Laptops

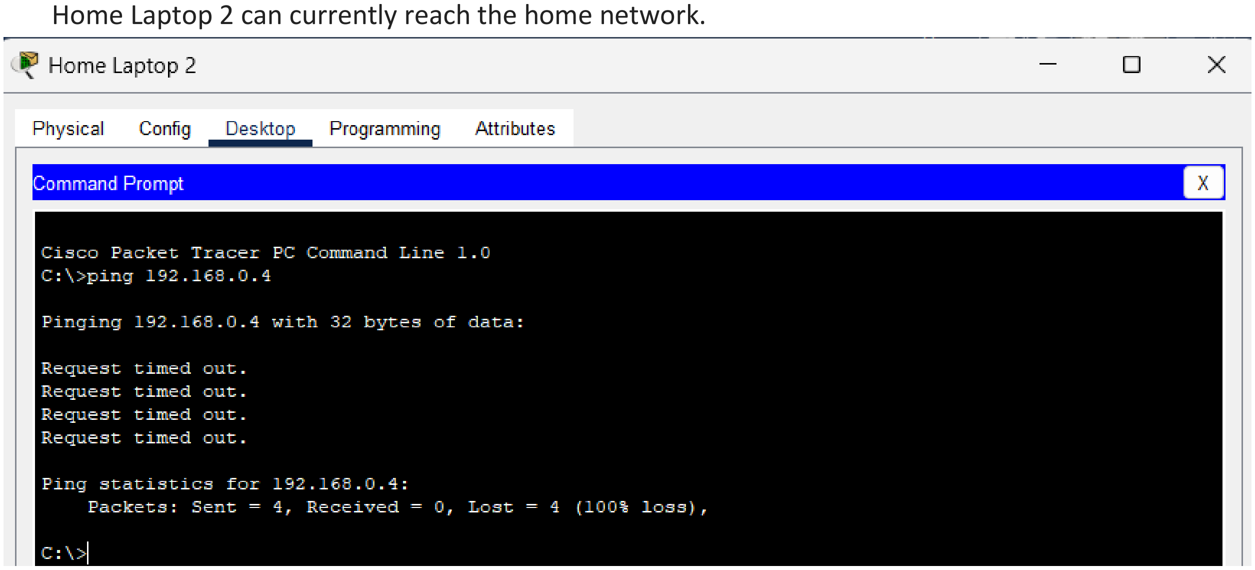
- **a.** Click Home Laptop 1 > Desktop tab > Web Browser.
- **b.** Navigate to www.ptsecurity.com. Click Fast Forward Time (Alt+D) to speed up convergence until the Data Center Public Web page loads.
- **c.** Repeat this step for Home Laptop 2.



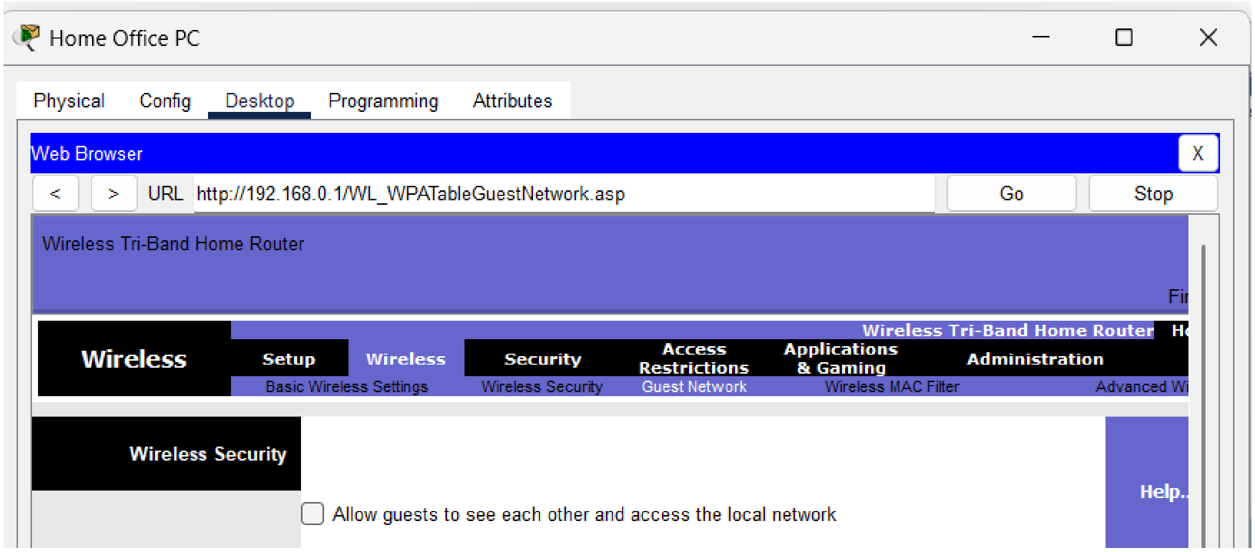
Step 2: Configure Security for GuestNet and HomeNet Interconnectivity

GuestNet and HomeNet should be isolated. This step confirms isolation is working and configures the router to enforce it.

- **a.** Use any method to determine the IP address of Home Laptop 1.
- **b.** Click Home Laptop 2 > Desktop tab > Command Prompt.
- **c.** Enter the ping command followed by Home Laptop 1's IP address. Note that pings succeed — Home Laptop 2 can currently reach the home network.



- **d.** From Home Office PC, log back into the Home Wireless Router at 192.168.0.1 if necessary.
- **e.** Click the Wireless tab, then the Guest Network submenu.
- **f.** Uncheck the box next to "Allow guests to see each other and access the local network", then click Save Settings.



- **g.** From Home Laptop 2, attempt to ping Home Laptop 1 again. The pings should now fail, confirming network isolation is enforced.

Conclusion

In this activity, a home wireless router was hardened by changing the default administrator password and disabling remote management. Wireless security was configured using WPA2 Personal with AES encryption for both the HomeNet and GuestNet networks. Wireless clients including laptops and IoT devices were successfully connected to the appropriate networks. Connectivity to the internet was verified for both laptops. Finally, guest network isolation was configured and verified, confirming that hosts on GuestNet cannot communicate with devices on HomeNet. These steps demonstrate key wireless hardening practices that reduce the attack surface of a home or small office wireless network.